

CS264 Assignment 3

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1 Submission Instructions

This assignment is **18%**, please upload your **source code** to moodle before 23:59, 12th-November. Please note that your source code has to be **C++**, otherwise your submission will be automatically rejected.

2 Late Submission

Late submissions will receive cumulative penalties to the awarded mark; 5% for first day, 15% for second day, 35% for third day, 65% for fourth day. Late submissions after the fourth day will not be accepted.

3 Plagiarism

Zero-tolerance policy with plagiarism is applied to this assignment. If your submission is found to be copied from someone else's work or machine generated or downloaded from internet. 0 mark will be given immediately and a case will be reported to the Department and University.

4 C++ Class

Design a C++ class called *Matrix* that can perform a set of basic matrix operations. Currently, this matrix only operates on integers.

4.1 Structure of the Class

You *must* use **dynamic memory allocation** to create your matrix, and safely free memory after using them. Your Matrix class must define the following operations (methods):

1. `Matrix(unsigned int m, unsigned int n)`

A specialised constructor that initialises a matrix of size $m \times n$, every element in this matrix is 0.

2. `Matrix(const Matrix &mat)`

A copy constructor that copies every element from matrix *mat*.

3. `Matrix(int** array, unsigned int m, unsigned int n)`

Initialise current matrix using a 2D array with size of $m \times n$.

4. `unsigned int rows() const`

Returns the number of rows in the matrix.

5. `unsigned int columns() const`

Returns the number of columns in the matrix.

6. `int get(unsigned i, unsigned j) const`

Returns the element at *ith* row and *jth* column. (*i* and *j* here begin at 0).

7. `void set(unsigned i, unsigned j, int value)`

Update value at *ith* row and *jth* column. (*i* and *j* here begin at 0).

8. `Matrix operator+(const Matrix &mat)`

Returns a matrix that contains the results of addition of two matrices.

9. `Matrix operator-(const Matrix &mat)`

Returns a matrix that contains the results of subtraction of two matrices.

10. `Matrix operator*(const Matrix &mat)`

Returns a matrix that contains the results of multiplication of two matrices.

11. `Matrix operator~() const`

Returns the transpose of current matrix.

12. `bool operator==(const Matrix &mat)`

Returns *true*(1) if two matrices are the same, otherwise *false*(0). Two matrices are the same iff they both contain the same element at exact location in each matrix.

13. `string toStr()`

Returns a string representation ((showing every element in the matrix)) of the matrix.

4.2 Comments

For each methods above, you should write a block of comment to explain what a method does including its return values and arguments (whether it is read-only or not). The comments will be **10%** of this assignment. Here is an example:

```
/**
 * check if two matrices are the same: two matrices are the same
 * if they both contain the same element at the same location in
 * each matrix.
 * @param mat is another matrix
 * @return true if both are the same, otherwise false.
 */
bool operator==(const Matrix &mat)
```

5 Test Cases

You should have at least one test case for each method above. Your test case should have the following format (with appropriate comments):

```
//test case 1 for matrix operation +.  
void Test_Case_1_Plus()
```

Test case is **30%** of the assignment.

6 Files Upload

You only need to submit 3 files:

1. **Matrix.h**: class definition of **Matrix**.
2. **Matrix.cpp**: implementation of class **Matrix**.
3. **TestMatrix.cpp**: test program for running your test cases. This is the file tutors are going to run.